

stand up to injustice and poverty. “This fight begins in your community,” he said. He spoke about the unity of the people on both sides of the border. Doyle’s address was inspiring, meant to motivate the Mexican schoolchildren to study hard to achieve their dreams. They received gifts from the department—plastic water bottles. Then they had a few minutes to change their clothes and get their towels ready before they got sprayed with water: an end-of-the-school-year tradition in Nogales. Delighted, they ran under the hose set up as a shower on top of Ladder 1, and screamed.

“They are all wetbacks now,” one of the firefighters said, looking at them.⁵¹

The contrast between the laughter of these Mexican kids and Araceli’s cry at the border fence was jarring to me. I needed a pause to comprehend what had happened only a few miles and one hour apart. To the firefighters in Nogales it was just another day at work. They could even joke about it.

TACTICAL INFRASTRUCTURE

“Tactical infrastructure,” as the Border Patrol defines it, is not limited to the fence.⁵² It includes gates, roads and bridges, drainage structures and grates, observation zones, boat ramps, and lighting and ancillary power systems, as well as remote video surveillance, which together “allow CBP to provide persistent impedance, access, and visibility, by making illicit cross-border activities, such as the funneling of illegal immigrants, terrorists, and terrorist weapons into our Nation, more difficult and time-consuming.”⁵³ Installed on the frontlines of the US government’s converging “wars” on drugs, terror, crime, and migrants, tactical infrastructure harms the latter category—people—most. A 2009 report for Congress, using the terms “tactical,” “operational,” and “deployed,” conveys the Border Patrol’s mission through the language of combat:

Border fencing is most effective for its *operational* purposes when *deployed* along urban areas. In these areas, individuals crossing the border have a short distance to cover before disappearing into neighborhoods; once they have entered neighborhoods it is much more difficult for USBP agents to identify and apprehend unauthorized aliens. Also, from populated areas it is relatively easy for unauthorized aliens to find transportation into the interior. For these reasons, all of the border fencing constructed by the USBP to date has been built in urban areas abutting the border, such as San Diego, Nogales, and El Paso. In rural areas, the USBP testified that it has a *tactical* advantage over border crossers because they must travel longer distances before reaching populated areas.⁵⁴

To determine the placement of tactical infrastructure, DHS devised an algorithm called the “border calculus.” It was part of the Secure Border Initiative (SBI), a multi-year, multi-billion-dollar program inaugurated in 2005 that combined the expansion of “tactical infrastructure” along more than 600 miles of the US-Mexico border with the creation of a “virtual fence”—a high-tech barrier consisting of surveillance towers that monitor activity and look for incursions using radar, sensors, and high-resolution cameras. Border calculus bolstered the program’s credibility by making security appear like science, emphasizing both the spatial and temporal dimensions of enforcement. In the words of Gregory L. Giddens, who directed the Secure Border Initiative, the chart lays out “a very simple algorithm that our ability to respond to a border incursion needs to be much less than the time it takes an illegal alien to get to a vanishing point.”⁵⁵ This is how he explained the meaning of the lines, dots, arrows, and circles in the chart:

The middle of the chart has borders, barriers, and fences along the port of entry. If you think about a border town such as Nogales, the time that someone would get to that vanishing point is very short, so you would want to use technical infrastructure to slow them down and allow us to have more response time. This basic border calculus chart and its governing algorithm that we want to make sure we can respond well within the time it takes an illegal alien to get to that vanishing point is what is going to guide us in a very systematic, disciplined manner to lay out the solution in each part of the border by understanding that it will change as we go forward. As we go forward in secure areas, the coyotes and the smugglers are going to react to that, and they are going to use different routes and different parts, and we need to be able to be less bureaucratic and more nimble in our approach so we can be more responsive to that so we can try to predict some of that, so we can be ready.

Temporality adds an important new layer to the conventional mapping of border security. But the chart flattens space (and time) into a two-dimensional system. It fails to account for the height and depth—the volume—of terrain.⁵⁶ The fence is not a line on a map that separates two planes. Movements of smugglers and migrants are not limited to the surface of the earth. In a three-dimensional space, they can breach a barrier by passing under it through a subterranean tunnel or using a ladder to climb over it. Initially, the border calculus missed the significance of terrain’s volume. But it took CBP only a few years to start embracing “the politics of verticality.”⁵⁷ Today, agents in Nogales send handheld drones to scout the tunnels extending under the international border, while from Fort Huachuca military base

southeast of Tucson, the agency operates unmanned aerial vehicles, including Predator B drones, equipped with radar and sophisticated sensors that can follow movement in real time and distinguish people from animals 25,000 feet below their flight path. Dubbed VADER (Vehicle and Dismount Exploitation Radar), these systems were developed for the US Army to detect enemy combatants planting roadside bombs in Afghanistan, but are now deployed to track men, women, and children who risk their lives for a chance to escape violence, reunite with their families, and join the exploited labor force.⁵⁸ Drones work in conjunction with checkpoints. Since 2007, a system of interior roadblocks has proliferated on all northbound roads in southern Arizona, providing visual testament to the extralegal powers that CBP claims within a zone that stretches 100 miles from the boundary of the United States.⁵⁹ Checkpoints link the horizontal and vertical axes of spatial border enforcement with temporality: they push unauthorized migrants off the main roads and into the desert, prolonging the “time to vanishing point” not by hours, but by days. “Vanishing” acquires a different meaning here: people disappear in the desert. Aerial surveillance—the eye in the sky—may detect those still moving against the clock on this pale canvas, before the bodily fluids that keep them alive run out. But others are gone without a trace. Vanished before they reach “vanishing point.”

A few days after Araceli was airlifted from Nogales and flown to Tucson, I drove north in hopes of finding her in recovery at the hospital. Two rescue helicopters stationed closest to the border, LifeLine 3 and LifeNet 6, can deliver a patient to the regional trauma center in about twenty minutes. It takes over an hour to cover the 70-mile distance by car. All ground transportation halts at a massive Border Patrol checkpoint on I-19 north of Tubac. Since its appearance as a “temporary” roadblock in 2007, undocumented border crossers and long-term residents alike have been avoiding calling 911. Going north to Tucson, where all major hospitals are located, requires them to pass the cameras, dogs, and agents at the checkpoint. If they seek medical care, they may be detained and deported. Paramedics who live and work in border communities understand this fear. Sometimes they can take undocumented patients to Holy Cross Hospital in Nogales, but it provides only basic services. Other times, no matter what they tell patients about the risk of refusing to go the hospital—they tell them that they could die—they still won’t risk going through the checkpoint.

Roadblocks annoy residents, who have to answer questions about their citizenship status whenever they leave their communities to go to a pharmacy or a grocery store. Outraged, they have staged demonstrations and monitored checkpoints on Sasabe Highway and on Arivaca Road.⁶⁰ Those who protest include liberal activists, ranchers, and libertarians, most of them white American citizens who feel secure to take a stand against practices that impinge on their freedoms and constitutional rights. However, the brunt of profiling falls on Mexicans and Mexican Americans. Alex explained how the checkpoint reinscribes ethnic hierarchies:

My sister's boyfriend is a typical white guy: he's big, he's tall, he's got white hair, he's bold, he's got a beard. You see him and you think you are looking at Santa Claus—typical white guy. He drove up to the checkpoint and they asked him if he had anything to declare. He said: "Yeah, I've got three handguns in the back and my shotgun." And the agent started laughing at him. He goes: "No, it's true. That's what I have." He opened the trunk and the agent looked inside and said: "OK, sir. Have a good day. 'Bye." He never asked him: "Are they yours? Why do you have them in there?" Nothing. "OK. 'Bye." That's all. When I go up there, they usually ask a bunch of questions: "Where are you from? What are you doing? Where are you going?" Sometimes they will send me to the secondary to wait there. And they'll bring the dog. Now even when I drive with the firefighter plates on they'll do that.

I rarely traveled with other people in the car, but, when I did, it was with firefighters from across the line. One time I accompanied three bomberos to pick up donated bunker gear and other supplies in Phoenix. We were riding in the command truck, a red Ford Expedition with "Nogales Bomberos" signs and a Mexican flag. As soon as we approached the checkpoint and started slowing down, the guys took their Border Crossing Cards out of their wallets and were ready to present them for inspection.⁶¹ The agent greeted us. He didn't ask whether we were citizens—probably Mexican plate numbers on an official vehicle made it clear. He only asked whether we had anything in the trailer that we were hauling behind. "Nothing, empty," Pablo, who was sitting with me in the back, said in English. "OK, have a good day," said the agent. Only later, when we made our first stop for gas, did I notice that the vehicle looked different that day: emergency lights on the roof were wrapped in a black cloth. Apparently, the CBP agent at the port of entry instructed the bomberos to cover up the lights so that Americans wouldn't confuse them for a "real" emergency vehicle. There are no official regulations that require Mexican rescue trucks to hide their

lights when in US territory, but some agents like to show who is in charge on the border.

As you approach the checkpoint, the speed limit gradually goes down from 75 to 15 . . . Signs command attention: “All vehicles must stop ahead.” “Use low-beam lights.” “K-9 on duty. Restrain your pets.” The agent waved me off before I could utter a greeting.

Beyond Tubac, the snaking highway straightens up, stretching out from Amado to Green Valley to Sahuarita. Vehicle crashes are frequent here—drivers become sleepy as they speed through the desert landscape, which looks the same, mile after mile. A vast crust of dry earth frying in the sun. Tall saguaros. Fanning ocotillos. Dust devils. Sometimes you can see smoke clouds rise from the Santa Rita Mountains. We were still in wildfire season, which starts in spring and ends with the first monsoon rains around the Fourth of July.

Dr. Hann met me in his office at the University of Arizona’s Department of Surgery, located on the hospital’s fifth floor. The border wall looms large in the emergency room where he works. “Most commonly we see people who are trying to cross the border by trying to climb the wall. In certain areas, the fence is up to 20 to 30 feet high. A fall from that height can be pretty serious. Very frequently we see patients with orthopedic injuries. Ankle fractures are very common, tib-fib—or lower extremity—fractures, and spinal fractures,” he said, listing the most common injuries caused by the barrier. “It’s a pattern of injury,” the surgeon explained. “When someone falls and lands on their feet, the energy is transferred from the feet all the way to the spine. We’ve had quite a few head injuries as well. With good outcomes. We’ve also had patients that had devastating head injuries who did not have good outcomes.” Dr. Hann has seen the limits of surgical intervention to saving lives on the frontlines—not just on the US-Mexico border; he experienced it in Haiti immediately after the earthquake in January 2010.

“You don’t know who is legal and who is illegal. You just treat them,” he told me. As trauma surgeons who work in a facility less than a hundred miles from Mexico, he and his colleagues get critically injured patients from both sides of the border as well as on both sides of the law. Undocumented border crossers are not the only ones receiving care here. There are Border Patrol agents who wreck their vehicles on the rugged desert terrain. There have been high-profile criminals from Mexico, including a woman who sustained a bad injury when she was shot in the head. Her recovery was difficult, remembers

Dr. Hann, but despite her family's protests she was repatriated to Mexico. The UMC Trauma Center got even more public exposure in December 2010, following an altercation between three Border Patrol agents and a group of heavily armed men—a “rip crew”—that robbed migrants and drug smugglers crossing through the canyons west of Rio Rico. Agent Brian Terry was wounded in the exchange of gunfire and died while his colleagues were rushing him to the hospital.⁶² On shift that day, Dr. Hann treated one of Terry's assailants, who was shot during the confrontation. “I think it was his liver, colon, and stomach . . . suffice to say, he had quite a few injuries.” The man lived and, after pleading guilty to first-degree murder, is serving a thirty-year sentence. Trauma surgeons don't always know the legal circumstances surrounding their patient's critical condition. They focus on what needs to be done to save a person's life: suturing arteries to stop the bleeding, lowering intracranial pressure to prevent brain herniation. “We don't care where they are from and what their nationality is, how much money they have. None of us cares about that. We just take care of them.”

The Emergency Medical Treatment and Active Labor Act of 1986 (EMTALA) requires ambulances and hospitals receiving Medicare funds to triage, treat, and stabilize any patient who seeks emergency services regardless of ability to pay or legal status. But injury and illness are not distributed evenly across the national territory, and neither are the costs of treatment. The deployment of tactical infrastructure on the US-Mexico border, which injures those who are undeterred by the government's security policies, has placed a considerable burden on health care providers in southern Arizona. Under Section 1011 of the Medicare Prescription Drug, Improvement, and Modernization Act of 2003, the federal government designated funds to compensate ambulance companies and hospitals for emergency care of undocumented migrants, but the reimbursement program ended in 2008.⁶³ However, the Border Patrol continues to bring about fifty patients with fall injuries to UMC every year.⁶⁴ UMC treats anyone who needs care, but most of the costs incurred by undocumented patients are written off because few are able to pay. Financial concerns have pushed the administration to seek ways of sending stabilized patients back to their country of origin. The hospital has an international patient services coordinator who works as a liaison with foreign consulates and helps locate adequate options abroad. Because so many of their patients have been from Mexico, over the years UMC has built relationships with health care providers in Sonora. Preparing a discharge plan for an unauthorized border crosser from Mexico is thus easier than sending

patients back to Honduras, El Salvador, or other countries that don't have established ties with Tucson-area hospitals. Dr. Hann:

From my standpoint, the frustrating thing is . . . you know with absolute certainty that the care that they get here is as good as anyone else's. The ones that go back to Mexico, you really wonder what's gonna happen to them. For example, patients with spinal fractures who need rehabilitation, they can hardly move. Once they go back to Mexico . . . I'm assuming most of these people who were trying to climb walls to get to the States don't have a lot of financial resources to begin with. Now they go back to Mexico in an even worse physical condition, having suffered pretty significant injury. What's gonna happen to them in terms of their ability to survive in their environment? What's their follow-up going to be? If there is a surgical complication, what's gonna happen to them? They are not gonna come back to the States to get treated. Say there's a hardware infection, what happens? I don't know. These are things that are unsettling, but at the same time it's . . . it's reality. Where does our care end?

I didn't find Araceli at the hospital. A woman at the information desk referred me to another woman in the administration who kindly agreed to look for her in the admissions log. But she was not there. Patients who arrive at the hospital through the Border Patrol are not registered under their real names. They stay at UMC for a few days and then they are taken to the Border Patrol station for processing. Some are detained and others are transferred to Nogales, Arizona, for immediate deportation. Mexican migration authorities meet them at the port of entry and transport those who need medical treatment to the Hospital General Nogales, where they stay until they are well enough to travel back to their hometown. Araceli could be anywhere now.

In 2015, the Mexican Consulate registered 59 Mexican nationals hospitalized in Tucson.⁶⁵ May was the busiest month: 11 cases. In 2016, they counted 66 people, 12 of them in March, 10 in July. Most of these were fractures caused by falling off the "muro fronterizo"—even the consulate calls it a "border wall," not a fence. Other common reasons for hospitalization included dehydration, injuries to the feet (blisters, cuts), and GI problems from drinking contaminated water. The consulate also registered bites by poisonous animals, spontaneous abortions due to severe dehydration, drowning in the arroyo during the rains, sexual abuse by human traffickers, and ingestion of cactus. The numbers may seem low, but that is because the consulate only learns about a patient when either the Border Patrol or the hospital lets them know. On the other side of the border, Juan Bosco migrant

shelter in Nogales, Sonora, accepted thirty injured people in June 2015 alone. That year the consulate and the migrant shelter both reported an increase in serious injuries along the fence.⁶⁶ Ricardo Pineda, the Mexican consul in Tucson, said that they were seeing more migrants in need of medical attention from falling off the border wall than from crossing the desert. Most of them were women with fractures to their feet, ankles, or legs. Gilda Felix, the director of Juan Bosco shelter, told the press: “They think it’s easier than walking for days in the desert but it’s not. [. . .] It’s the same crossing through the wall or through the desert, both difficult and dangerous.”⁶⁷

Several weeks went by before I saw JLo, the paramedic who treated Araceli, again. He heard that a woman in Nogales, Sonora, complained on the radio about the ill treatment by the Border Patrol. She said that when she fell off the fence, she was left lying there for hours. She said she saw Border Patrol agents passing through, but they didn’t stop to help her. JLo was sure it was Araceli. “It seems she has been just deported back to Mexico,” he told me. Shortly afterward, news media in Arizona circulated a story about a thirty-one-year-old woman from Oaxaca who spent twenty days recovering from injuries to her legs and spine at Juan Bosco shelter for migrants in Nogales, Sonora.⁶⁸ She told the reporters that she paid smugglers \$13 to climb a ladder propped against the fence, but once she reached the top, she didn’t know how to get down and slipped. A Border Patrol agent spotted her two hours later. She broke both of her legs and her spine. With the assistance of the Mexican Consulate she was being flown back to Oaxaca, where she was looking at a six-month recovery before she could walk again. Her injuries meant that she could no longer work to pay for the needed therapy and continued medical care, the newspaper noted matter-of-factly.

It may or may not have been Araceli. Hers is but one among many painful stories echoing back and forth across a militarized border.

In the middle of the nineteenth century what is now the steel wall separating Mexico and the United States in Nogales was just a small pile of rocks, “a pyramidal monument of dressed stone.”⁶⁹ It was put in place by the Joint United States and Mexican Boundary Commission, sent to survey and map the region following the US-Mexican War. Lieutenant of Topographical Engineers Nathaniel Michler described “the pretty little valley of Los Nogales [. . .] clothed with rich green verdure,” contrasting it with “the bleached barrenness of the Colorado and Gila” that they encountered to the west. In the 1890s, when the

US-Mexico boundary commission sent the Corps of Engineers back to the line, the pyramid of stones left in Nogales during the first survey was replaced by a 6-foot tall four-sided obelisk and renamed Boundary Monument 122.⁷⁰

The origins of the fence that divide Nogales, Arizona, from Nogales, Sonora, may surprise many. The historical record shows that it was first proposed to protect Mexicans. Due to their strategic location, during the Mexican Revolution border towns became battlefields. Stray bullets that crossed the boundary line killed a handful of Americans. Rising anxieties about security, amplified by refugees seeking shelter in the United States, warranted the placement of National Guard troops on the border. There was no physical barrier between the two sides: the bodies of armed soldiers walking along the boundary enacted the division that was otherwise unmarked. According to historian Rachel St John, the idea to construct a 6-foot wire fence was first raised by the presidente municipal of Nogales, Sonora.⁷¹ Mexican authorities hoped that it would prevent US soldiers from accidentally shooting Mexican people. The mayor didn't survive the Third Battle of Nogales, but the US government followed through with his plan and extended a single strand of barbed wire separating the two cities. In 1929, it was swapped for a 6-foot chain-link fence, which was later doubled in height downtown. This fence remained the material form of the border for most of the twentieth century. Still today, residents nostalgically remember the so-called "picket fence" between neighbors.⁷² It served as a marker of the international boundary for legal purposes, but it let the sights and sounds get through. The fence was easily disassembled during parades on official American and Mexican holidays, allowing ties between the communities to be maintained.

William Sanchez, assistant chief of the Nogales Fire Department, remembers that unauthorized crossing was very common. "When it used to be the old fence, people would cross, go in and out. Not a big thing." Many of those who worked in the fire service were local, and over the years they experienced changes to border infrastructure as residents of these towns. "When we were kids, we used to cut through the fence and go buy bread and stuff over there [in Mexico] and then come back," another firefighter told me. Another reminisced about a man on a bicycle who would bring lemons and cheese from Nogales, Sonora, to sell them at the fire station in Nogales, Arizona. The fence was there, but it didn't mean much to anybody.

However, in the mid-1990s, Operation Safeguard brought to Nogales the Border Patrol's "prevention through deterrence" strategy. The new metal wall separating Mexican and American urban spaces was made from surplus steel

planks that the US military used as portable pads for landing Hercules cargo planes and Huey helicopters during the Vietnam War.⁷³ Even the US Army Corps of Engineers, who designed the M8A1-style solid corrugated steel panels, acknowledged the flaws of their construction: the landing pads had rough edges that frequently ripped the tires of heavy aircraft. By the end of the war the military had replaced them with aluminum mats, relegating M8A1s to taxiways and parking lots.⁷⁴ After Vietnam, landing mats were easily repurposed for other ends. Measuring 12 feet long, 20 inches wide, and a quarter-inch thick, and weighing 147 pounds, the panels became free “army surplus” and were redeployed to the US southwest border.⁷⁵ Each mile of fence required 3,080 metal sheets.⁷⁶ By 2006, they were used to build over 60 miles of border fence in California, Arizona, and Texas. In Nogales, the new rusty steel barrier was first erected in 1994. In the beginning, it stretched for nearly a mile, but was soon extended to 2.8 miles. Depending on the location, the landing mat fence was 8 to 12 feet high, and it was “enhanced” with features to make climbing over it more difficult. The community didn’t like it, as even Commissioner Meissner admitted. “We would love to build more aesthetic looking fences,” she said, but it would be “extraordinarily more expensive.”⁷⁷ Despite their initial enthusiasm with free construction materials, the Border Patrol soon noticed the drawbacks too. “The landing mat fence, it is opaque,” Chief Kevin Stevens said in a congressional hearing.⁷⁸ The solid texture provides border crossers with tactical advantage. “They have time. They have time sitting on the foreign side to be able to attempt to defeat it.”

In 2011, the landing mat wall was removed to build a sturdier and taller bollard-style barrier, made of rectangular steel tubes reinforced with concrete, with metal plates on top. Extending up to 20 feet above the ground and 10 feet below the surface, this wall was designed to act as a more effective deterrent against climbing over or digging under.⁷⁹ See-through fence was also safer for the Border Patrol: officers could anticipate “rocking” and, a year after the fence was completed, assaults on agents were down by over 60 percent.⁸⁰ Even residents were reportedly happier with the face-lift: 4-inch gaps between bars allowed them to hear mariachi music, watch neighborhood movement, and chat with their neighbors.⁸¹ At least that’s what was celebrated in the press.

If you talk to the locals, not many would say they like the barrier. On the Mexican side, it went from being known as *el cerco* (the fence) to *el muro* (the wall).⁸² In October 2015, Tijuana artist Ana Teresa Fernández enlisted bystanders to help her paint a stretch of the fence west of the Dennis DeConcini port of entry in Nogales, Sonora. In her performance, rusty dark

brown bars turned blue, merging with the color of the sky and becoming “invisible.” The project was called “Borrando la frontera”—erasing the border. One of the men who picked up a brush, dipped it into a bucket of blue paint, and worked alongside Fernández was a deportee separated from his family in the United States; another was a Mexican police officer.⁸³ Dozens of volunteers joined the effort to create the optical illusion that the wall had “disappeared.” But the art project didn’t change anything. A Nogales resident told a news reporter: “I wish it was magic like that, but you can’t just erase it.”⁸⁴

“When they first built the fence, I worked for Nogales Fire and we had a lot of people jumping the fence and breaking their ankles,” remembers a former city paramedic who is now a captain in a district further north of the border. The same sharp edges that damaged the tires of the military fleet in Vietnam caused large gashes, limb amputations, and degloving injuries to the people who were scaling the barrier separating them from jobs at construction sites, hotel kitchens, and vegetable fields in the north.⁸⁵ “When they were climbing down, [border crossers] would slip and their hands would get stuck up there, so they would get their fingers cut off,” is how Alex described it.

The border sliced those who challenged it, cruelly mutilating the bodies of men—it was mostly men back then—who were coming to find work. “They would land on this side, and the finger parts would land on the other side, usually; or they would get lost between the plates, so we could never get them out,” Alex recalls. They had to tell the injured people that their fingers were gone. “I don’t know how many times we searched and searched and searched and never found the fingers.” He continued: “In some places they used to have openings at the bottom [of the fence] with grates on them for the water to go through. You could still see across and you could see the fingers [. . .] on the other side of the border, and the people were over here. Sometimes we would reach over and grab the body part, [. . .] and put it on ice.” Other times people on the other side would pick up the fingers and hand them over the fence. “We would use trauma dressings, 4x4s, and lots of ice packs.”

The Border Patrol knew about this. Chief Stevens, who oversaw the Nogales station in the early 2000s, mentioned the injuries when he spoke to the US Congress:

People were climbing over the top of the landing mat fence depending on the height of it or, in some cases, they would put ladders up against it and come

over the top. One of the things I experienced when I was in Nogales was people who were not really physically capable of climbing the fence on their own, they would get assistance to climb the fence and then not be able to handle their own weight when they came over the top of the fence and we would have people losing fingers on the fence, we would have people breaking ankles coming to the ground with compound fractures. That was among the things that we were faced [with] that prompted us to place an additional structure on top of the fence that even with assistance, somebody who was not physically strong would not be able to negotiate the fence. It stopped those people from even trying and significantly delayed even the most able.⁸⁶

“I thought it was just inhumane,” Alex said about those dispatches to the border fence. “One time we went to a call where the guy cut off his finger. I think it was his index finger or his pinkie . . . I can’t remember which finger it was. One guy was stabilizing him and the other guys were looking for the finger. And we saw a piece of his finger being carried away by ants. The ants went into Mexico and they went into the ant hole with a piece of his finger.”

I was warned not to take stories like these for granted. In a conference room in Washington, DC, where I gave a presentation about the tactics of security and the gory details of trauma to an audience of anthropologists, colleagues criticized me for believing these narratives, which they said were exaggerated. Firefighters always boast about their calls, they told me. I knew that rescue stories can become heroic tales, with each repetition further removed from reality. Back on the border later that spring, Alex and Bojo, who responded to that call when the ants carried a piece of a finger into an ant hole in Mexico, shrugged their shoulders, and said: “But that’s what happened.” They reached a consensus that it was the pinkie. The genre of everyday life on the line was best described as magical realism. It didn’t have to sound reasonable to be true.

In Douglas, about two hours east of Nogales, paramedics have also treated numerous trauma patients. “We are not seeing just closed fractures. You see the bone sticking out,” said Gerardo, who became a Cruz Roja volunteer in Agua Prieta at the age of fourteen and has been working for Douglas Fire since 2009. His colleague Jorge, who also began in the Cruz Roja in Mexico and later joined the fire department on the US side of the border, said it was actually worse “before they built the big fence.” But wounds didn’t stop when the new barrier came up: “We had a lot of falls from the fence. We had head traumas, back injuries, pelvic fractures.” They often ended up going to “ankle alley” two or three times a day.

Over the years, the changing design of the barrier has produced different types of wounds. While the earlier version, made of corrugated sheet metal, caused gashes and amputations, the present border wall is difficult to hold on to and results in multi-system trauma. Most commonly, these are ankle, tib-fib, or femur fractures. Once, when Nogales Fire Department was tied up working a structure fire, two paramedics from Rio Rico came to provide mutual aid in the city: “We went to the border. A lady had paid to go up a ladder. She decided she didn’t want to jump, and, whoever she paid pushed her off the fence,” one of the paramedics told me. “There’s a good 10 feet of jagged rocks near the fence. When she fell, her ankle was so severely broken it was almost amputated from her foot. She had tib-fib fracture, possible femur fracture, and her foot was literally separated. She couldn’t even feel it. When I put my hand to palpate, I could almost put my fingers through, I could see my bloody glove.”

Calls to help the injured along the border fence could be dangerous. While the Rio Rico paramedics were trying to stabilize the injured woman—start an IV and splint her foot—they heard something hit the steel barrier. At first, they thought it was gunshots, but it turned out somebody was throwing rocks. Border Patrol agents who were with the paramedics ran to hide behind their vehicles. “It’s almost like a war zone. You have the good and bad fighting against each other, but you are literally in the middle, trying to take care of the patient. And we did,” the paramedic said. Such incidents were not frequent, but they did happen. “We secured her foot as best we could with a pillow splint because it was about to fall off her body. We did our C-spine precautions because she was pushed over 20 feet.” When they took the patient to Holy Cross, she was immediately airlifted to the regional trauma center.

Few knew these dangers better than JLo. As a paramedic, he is regularly dispatched to help people at the fence. He looks petite in the company of his peers and punctuates his sentences with “neta” and “güey.” JLo is careful and attentive to detail—during downtime between calls he sits at the dining table with a tiny screwdriver replacing damaged iPhone parts, a side job to help pay the bills. When on scene, he acts fast. Once we were called to a residential apartment community to check on a woman who complained of high blood pressure. As we were rushing to wrap the cuff around her arm and take her vitals, Frank jumped and shouted: “Un ratón!” There was a mouse. JLo immediately turned to the patient: “Is it your pet, señora?” When she shook her head, he straightforwardly asked: “Do you want us to kill it for you?” She

didn't seem to object and JLo crushed the mouse with his boot. It was not the first time this happened. In the firehouse, the guys used to call him Percy, after a character from the movie *The Green Mile*, the one who killed the pet rodent Mr. Jingles in the prison.

JLo didn't plan on becoming a firefighter. He wanted to join Border Patrol. With a criminal justice degree from Pima Community College in Tucson, his chances of being hired by the federal agency were fair. But when the twenty-one-year-old went in for a medical exam, he was told he had poor vision and needed surgery. "I was really disappointed," he remembers. Thinking that he was too young to undergo vision correction, JLo accepted an invitation from his brother-in-law to volunteer at the Nogales Suburban Fire District. He became an EMT, started at the fire academy in Tubac, and two months before he got his certificates, JLo was offered a job at the Nogales Fire Department. It was 2002. Six years later he finished paramedic school and, after a brief period working in Tucson, decided to return and stay in the border town.

Some stretches of the wall are more perilous than others. In downtown Nogales, there is an offset landing with cement and rocks, where most dramatic injuries occur. "That's probably about a 4- to 5-foot area, like a sidewalk, but with rocks sticking out from it," JLo said. We were having chilaquiles and menudo for breakfast at El Zape, a Mexican restaurant right in front of the fire department headquarters. "What other reason are these there for?" Bojo chimed in, referring to the rocks near the fence. "They are there to injure people so that they couldn't run from the Border Patrol," he said. The firefighters knew this from experience, having responded to help numerous patients with orthopedic, spinal, and head trauma on that small piece of land right next to the CBP parking lot.

"We had people that landed on their head and died," one of them said. In February 2012, a forty-four-year-old man from Oaxaca, Mexico, died on the west side of town when he sustained head and neck injuries. Two years later, in March 2014, a forty-one-year-old man from El Salvador died from head trauma after falling from the border fence near the end of Short Street.⁸⁷ In 2016, the Nogales Police Department was investigating the death of a thirty-two-year-old woman from Juchitepec in the state of Mexico whose body was found near the border wall east of Nogales. She, too, had fatal injuries after possibly falling off the fence, the police said. Security buildup on the border was largely based on weaponizing the already difficult physical terrain. To emergency responders the causation was obvious: they did not hesitate to implicate the agency in deliberately altering the landscape to cause



A strip of asphalt with rocks along the border fence in Nogales, Arizona.

serious injuries to those who did not have the required documents to cross through the designated port of entry.

The fence constitutes what EMTs call an MOI, “mechanism of injury”: the forces that act on the body to cause damage. Protocols regulating prehospital medical services list the conditions under which emergency responders must transport patients to a major trauma center, often by air.⁸⁸ It includes anatomic and physiologic criteria, such as penetrating injury, amputation of a hand or a leg, fractures of two or more long bones, or a Glasgow Coma Scale of less than 14. But even when these signs and symptoms are not evident, falls from heights greater than 20 feet for adults and two times their height for children—that is, any falls from the border wall in Nogales—are severe enough to raise the index of suspicion for life-threatening blunt trauma: skull fracture, tension pneumothorax, internal bleeding. This mechanism of injury alone warrants immediately transporting the patient to a hospital with a surgical unit. Such patients are placed in a helicopter that passes above the checkpoints and lands in Tucson less than half an hour later. The Border Patrol and the Mexican Consulate decide what happens when they are stable and can be released. From the medical perspective, the current bollard wall in Nogales is thus worse than the previous corrugated steel fence: finger amputations did not trigger trauma alert. Those wounds were bandaged at the local hospital, and an hour or two later the ambulance would return the injured to the line.

POR OTRO LADO

In Nogales, Sonora, a few blocks west of the port of entry on Calle Internacional, there is a building covered with bullet holes. They date from the night of October 10, 2012, when Border Patrol agent Lonnie Swartz, standing on the US soil, aimed his gun through a gap in the fence and shot sixteen-year-old José Antonio Elena Rodríguez. The Border Patrol accused José Antonio of throwing rocks at the agents and police officers who responded to a call about suspected smuggling. The autopsy showed that ten gunshot wounds were angled from the back to the front: the teenage boy was hit from behind. A federal grand jury indicted Swartz in 2015, but the trial has been delayed several times.⁸⁹ Since the shooting happened through the fence, with the Border Patrol agent standing in one country and the victim in another, the case presents a legal limbo. In a similar cross-border shooting in